

MONCHO UNSCHOOLING

Volcanoes

Unit Study

“What if you could stand at the edge of the Earth's biggest secret and learn to read the mountain that breathes fire, ash and new land?”

Earth Science | Chemistry | Geography | History | Math | Language Arts | Art & Creativity | Life Skills & SEL | Movement & Body | Astronomy

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🧭 **How to use this guide:** This is not a workbook, and there's no answer key — on purpose! Every challenge is an invitation. Pick the ones that spark curiosity, skip the rest, and go find the answers by doing. **AI plans. Kids DO.** 🚀

🌋 EARTH SCIENCE

Volcanoes come in different shapes and made of different rocks, and you can learn to read all of it with your own hands.

🏔️ Challenge 1: Two Mountains, Two Slopes

🎯 Comparing shapes

Why would two volcanoes made of the same fiery stuff end up with completely different shapes? ❓

Mauna Loa in Hawaii is a shield volcano, with gentle, wide slopes. Mount Fuji in Japan is a stratovolcano, with steeper, cone-shaped slopes built from layers of ash and lava. Build them and see the difference yourself.

1. Mix 2 cups of flour, 1 cup of salt and about 3/4 cup of water into a dough (or use modeling clay). Split it in half.
2. On a tray, shape one half into a low, wide mound, like an upside-down shallow bowl. This is your shield volcano.
3. Shape the other half into a tall cone with steep sides. This is your stratovolcano.
4. Lay a ruler on the tray beside each model. Measure how tall each one is and how wide it is across the bottom. Use the same unit (all centimeters or all inches) for every measurement.
5. Divide the height by the width for each model. The bigger number means the steeper slope.
6. Make two small cards that say "Mauna Loa" and "Mount Fuji" and set each one next to its matching model.

Why: Runny lava flows far before it hardens, so it spreads into a wide, gentle shield. Thicker lava and ash pile up close to the vent, so they stack into a steep cone.

🌱 **Make it Easier:** Skip the division. Just look at the two models side by side and say which one is steeper and which one is wider.

🚀 **Level it Up:** Add a third model of a volcano with a slope halfway between the two, and predict where it would fall on your height-divided-by-width scale before you measure it.

🧪 Challenge 2: Float or Sink?

🎯 Predicting and testing

How can a rock that came out of a fiery volcano float like a piece of wood?❓

You're the rock lab technician today. Your job is to test a rock and write up what you find. Pumice is a volcanic rock so full of gas bubbles that many pieces float on water.

1. Collect a pumice stone (a beauty-supply store or pharmacy often sells one for foot care) and an ordinary rock of similar size from outside.
2. Draw a chart with two columns, one for each rock, and write your float-or-sink prediction for each.
3. Fill a bowl with water and gently lower the pumice in. Then do the same with the ordinary rock. Record what happens.
4. Look closely at the surface of each rock with a magnifier. Count or estimate the tiny bubble holes you can see.
5. Sketch what you see in each column of your chart, next to your prediction.
6. If your pumice sank, check it again after a minute. Water can slowly fill the holes.

Why: The holes in pumice are pockets where gas bubbles were trapped as the rock hardened. Those holes are full of air, which makes the whole rock light enough to float.

🌱 **Make it Easier:** Skip the counting. Just test both rocks and draw one big circle showing what the holes look like on the pumice.

🚀 **Level it Up:** Test two or three more rocks from around your home or yard. Rank all of them from most likely to float to least likely before testing, then check how good your predictions were.

🪨 Challenge 3: The Rock Family Walk

🎯 Classifying rocks

How can you tell which rocks from your neighborhood once came from inside a volcano?❓

Basalt is a dark, common volcanic rock that makes up much of the ocean floor and the islands of Hawaii and Iceland. If you can find a piece of basalt, study it first. If not, look up photos of Hawaii's black rocks.

1. Look at your basalt (or the photos). Note its color, how heavy it feels, and any tiny holes or grains.
2. Take a bag outside and collect 8-10 different rocks from a walk, a park or your yard.
3. Spread them out on a tray and compare each one to basalt. Does it look dark and fine-grained, or does it look different?
4. Sort the rocks into two groups: "looks volcanic" and "does not look volcanic."
5. Write a name card for each rock describing its color and texture, and set each card by its rock on the tray.

🌱 **Make it Easier:** Collect just 4-5 rocks and sort them by color only, dark and light, before comparing them to the basalt.

🚀 **Level it Up:** Add a third group called "not sure" and use a magnifier to look for clues, such as tiny holes or sparkly grains, that might move each rock into one of the other groups.

🧪 CHEMISTRY

Volcanoes are giant chemistry labs where gas, melted rock and minerals mix, cool and change form.

🥤 Challenge 1: Shake the Fizz!

🎯 Dissolved gases

Why would a bottle that is cold hold on to its bubbles differently than one that is warm? ❓

1. Chill one sealed bottle of fizzy soda in the fridge overnight. Leave an identical sealed bottle in a warm spot, like a sunny windowsill, for an hour.
2. Go outside with an adult, wearing old clothes. Shake both bottles for the same count of 20 seconds.
3. Open the cold bottle first, turning the cap very slowly. Watch how much foam rushes out.
4. Repeat with the warm bottle, turning the cap just as slowly. Which one fizzed more?
5. Draw both bottles and write one sentence explaining the difference.

Why: Gas stays dissolved in liquid better when it is cold. Warm liquid lets gas escape fast, and gas bursting out of a trapped space is what makes explosive eruptions so powerful.

🌱 **Make it Easier:** Use just one bottle. Shake it, open it slowly outdoors, and draw what the foam did.

🚀 **Level it Up:** Add a third bottle that stays at room temperature. Predict where it will land between the other two, then test your prediction.

🍬 Challenge 2: Sugar Obsidian

🎯 States of matter

How can the same melted stuff turn into totally different solids just by how fast it cools? ❓

You're the glass-maker. Obsidian is natural volcanic glass that forms when lava cools very fast. Ancient peoples chipped it into very sharp blades and arrowheads. You'll make a sugar version.

1. With an adult at the stove, melt 1/2 cup of sugar in a dry pan over medium heat, stirring gently until it turns clear and golden. Be careful, because melted sugar is extremely hot.
2. Have the adult pour it in a thin layer onto a cold tray (chill it in the freezer first).
3. Let it cool completely. Do not touch until it is hard.
4. Tap your sugar glass with a spoon. Then put a few grains of regular sugar beside it and look at both with a magnifier. Which one looks like smooth glass? Which one has tiny crystal shapes?
5. Label your samples "fast-cooled glass" and "slow-grown crystals."

Why: Cooling very fast freezes the atoms in place before they can arrange into crystals, which makes glass. Cooling slowly gives them time to line up into crystals.

🌱 **Make it Easier:** Skip the magnifier. Just tap your sugar glass and describe what it looks and sounds like.

🚀 **Level it Up:** Hold each sample up to a window and compare how much light passes through. Draw what you see.

🌋 Challenge 3: Lava Flow Race

🎯 Viscosity

Why do some lava flows creep slowly while others race down a mountain? ❓

Magma is melted rock underground, and the same molten rock is called lava once it reaches the surface. You'll test how thickness changes the way it travels.

1. Mix 1/2 cup of flour with 1/2 cup of cooking oil in a bowl for a runny lava. In another bowl, mix 1 cup of flour with 1/2 cup of oil for a thick lava.
2. Prop a baking tray on a book so it makes a gentle slope. Mark the starting line with tape.
3. Pour the runny lava at the top and start counting seconds. Stop when it quits moving.
4. Wipe the tray, then pour the thick lava from the same spot.
5. Measure how far each one flowed with a ruler and take a photo of each.

Why: Thick, sticky liquid resists flowing, so it moves slowly and piles up. Runny liquid slides easily and spreads far, just like different kinds of real lava.

🌱 **Make it Easier:** Pour the two mixtures one after the other and just say which one went farther, no measuring needed.

🚀 **Level it Up:** Make a third mixture with even more flour and tilt the tray steeper. Predict the order of the three flows before you pour.

🌍 GEOGRAPHY

Volcanoes are not scattered at random, and learning to read a map is how you spot where the Earth is most alive.

🗺️ Challenge 1: Trace the Ring of Fire

🎯 Map reading

If you connected the world's most famous volcanoes with a line, what shape would appear? ❓

You're the cartographer, and the map is blank.

1. Print or draw a world map with the Pacific Ocean in the middle.
2. Use an atlas or a globe to find the volcanoes named in this unit. Stick a sticker on each one.
3. Pick your brightest color and trace a line that follows the edge of the Pacific Ocean, through the stickers. You're drawing the Ring of Fire, a horseshoe-shaped belt around the Pacific where roughly three quarters of the world's active volcanoes are found.
4. Color the Ring of Fire so it stands out, then add a title and a compass rose.

🌱 **Make it Easier:** Use a map that already has the Pacific labeled, and mark only five volcanoes.

🚀 **Level it Up:** Add a second color for volcanoes that sit far from the horseshoe, and find out why they are there.

🏝️ Challenge 2: The Island That Rose From the Sea

🎯 Observing change

Can an island be born in a single lifetime? ❓

In 1963, near Iceland, an eruption began under the sea and kept going until a brand-new island appeared. It's called Surtsey, and scientists have watched plants and animals arrive there ever since.

1. Fold three sheets of paper in half to make a little book.
2. Page 1: draw an empty, open sea with nothing there but waves.
3. Page 2: draw the island growing, with smoke, ash and glowing rock rising from the water.
4. Page 3: draw the island as it is today, with the plants and birds that have arrived. Think about how a seed or a bird could reach an island far from land.
5. Give your book a title and a cover.

🌱 **Make it Easier:** Draw only two pictures, before the island and after it.

🚀 **Level it Up:** Add a fourth page showing the island 100 years from now, and write one sentence to explain what you drew.

🍇 Challenge 3: Would You Farm on Etna?

🎯 Landscape study

Why would anyone choose to live on the side of a volcano? ❓

You're going to fly to Mount Etna, on the island of Sicily.

1. Open a free map (OpenStreetMap or Google Earth) and fly to Mount Etna. Zoom in and count the villages, farms, vineyards and orchards on its slopes. Trace a farm road with your finger.
2. Go outside and dig up a cup of soil from your garden or a park. Rub it between your fingers. Look for gritty bits and tiny shiny specks, and sort what you find into small piles.
3. Compare your soil with Etna's. Think about which one you would rather plant in.
4. Draw a map of Etna's slopes. Put a star where you would build your house and label what you would grow.
5. Pour your soil sample into a jar and label it with the date and where you found it.

Why: Ash and lava slowly break down into minerals, so volcanic soil is often very fertile. That's why farmers still grow crops on volcano slopes.

🌱 **Make it Easier:** Count only the vineyards on the map and sort your soil into two piles, gritty and smooth.

🚀 **Level it Up:** Find a second volcano where people farm, such as one in Indonesia, and compare it with Etna. Which one would you choose?

📜 HISTORY

Long before we had cameras and satellites, people wrote down, painted and retold what volcanoes did to their lives, and those records are how we know the past.

📜 Challenge 1: Dig Up Pompeii

🎯 Archaeology

How can a buried town tell us what daily life was like 2,000 years ago? ❓

You're the archaeologist at Pompeii. In 79 CE, Mount Vesuvius in Italy erupted and buried the Roman towns of Pompeii and Herculaneum under ash and pyroclastic flows. The ruins can still be visited and searched online.

1. Look at photos of Pompeii's plaster casts and frescoes. What do they show about the people who lived there?
2. Gather 5 small household objects (a spoon, a button, a coin, a bottle cap, a toy). Bury them in a tray of sand while a family member looks away.
3. Hand your partner a paintbrush and let them "excavate" slowly, layer by layer.
4. For each find, write in a notebook: what it is, how deep it was, and which corner of the tray it came from.
5. Guess how each object might have been used by its owner.

🌱 **Make it Easier:** Bury only 3 large objects near the surface and just list their names.

🚀 **Level it Up:** Divide the tray into a grid (A1, B2...) with string and record each object's exact square, then draw a map of the dig.

⚖️ Challenge 2: News Race

🎯 Debate and Communication

When a volcano explodes on the other side of the world, how fast should the news reach you? ❓

In 1883 the Krakatoa explosion was heard thousands of kilometers away. Today, news of an eruption could be on your phone in minutes. Split into two teams, one for "1883" and one for "today," and debate: how did news travel then, and how would it travel now?

1. The 1883 team researches the telegraph: how it sent messages through wires and what slowed news down.
2. The today team lists every way news travels now: phones, TV, the internet, satellites.
3. Each team gets 2 minutes to present its best arguments, then 1 minute to answer the other side.
4. Make a scorecard with columns for Speed, Reach and Trust. Give each team 1-5 points in each column and write the winning reasons.

🌱 **Make it Easier:** Skip the research and debate with only two reasons per team.

🚀 **Level it Up:** Add a third team, "messenger on horseback," and argue what is lost and gained when news is slower.

💌 Challenge 3: Remember Armero

🎯 Writing with Empathy

How can a few words and a drawing keep a place alive in people's memory? ❓

Lahars are fast mudflows of volcanic ash and water. In 1985, a lahar from Nevado del Ruiz in Colombia buried the town of Armero.

1. With an adult, read a child-level account of what happened in Armero. Notice who lived there and what the town was like.
2. Find out what people learned afterward about warnings and being ready.
3. Fold a piece of paper in half to make a card.
4. On the front, draw something that represents Armero, like its streets, its people or the mountain.
5. Inside, write 2-3 sentences that remember the town and say what was learned.

🌱 **Make it Easier:** Write just one sentence and draw a simple picture.

🚀 **Level it Up:** Write a short poem or letter from a child in Armero today to a child who lived there in 1985.

➗ MATH

Volcanoes are full of big numbers, and math is how we shrink them down, compare them, and read their shapes.

🏔️ Challenge 1: Shrink a Giant Volcano

🎯 Scale and Measurement

If the biggest volcano on Earth were squeezed into your hallway, how would it compare to a door or a tree? ❓

Mauna Loa in Hawaii is the largest active volcano on Earth by volume, rising about 4,170 m above sea level, and much higher from the sea floor. You will shrink that height so it fits in your home.

1. Pick a scale: every 1,000 m of volcano becomes 1 m on your tape. How tall is Mauna Loa now? (Hint: 4,170 ÷ 1,000.)
2. Stretch a measuring tape along a hallway wall and mark that height with masking tape or a sticky note.
3. Measure a door (about 2 m), a tall houseplant or the tallest person at home, and mark each one beside the volcano line.
4. Label each mark and write how many doors tall your mini Mauna Loa is.

🌱 **Make it Easier:** Use a scale of 1 m for every 1,000 m and round the volcano to 4 m. Just compare it to one door.

🚀 **Level it Up:** Add a second scale where 1 cm equals 100 m. How long would the line be now, and does it still fit in your house?

🌽 Challenge 2: How Fast Did a Volcano Grow?

🎯 Rates and Division

If a mountain pushed up out of a cornfield, how much taller did it get every single day? ❓

You're the volcano tracker. In 1943, Paricutín in Mexico began as a crack in a farmer's cornfield. In its first year it grew a cone about 336 m tall, and it kept erupting until 1952. Time to work out its growth rate.

1. Divide 336 m by 12 months. How many meters did it grow per month?
2. Divide 336 m by 52 weeks. How many meters per week? (Round to the nearest whole number.)
3. Divide 336 m by 365 days. About how many meters per day? (Estimate with friendly numbers, like 365 ÷ 365.)
4. Make a bar from pennies or building blocks where each one stands for 1 m. Line up the daily growth and compare it to the weekly and monthly bars.

Why: Eruptions pile up ash and cinders around the vent, so the cone grows a little with every blast, building the mountain on the spot.

🌱 **Make it Easier:** Only work out meters per month, and round the height to 340 m and the year to 10 months to keep the numbers friendly.

🚀 **Level it Up:** Paricutín erupted until 1952. If it had kept growing at your daily rate for 9 years, how tall would it have been, and why is that probably wrong?

📐 Challenge 3: Steep or Gentle?

🎯 Angles and Shapes

How can the angle of a paper triangle tell you what kind of volcano it is? ❓

1. Cut two triangles from paper: one tall and skinny, one short and wide. Each is the side view of a volcano.
2. Place your protractor along the bottom edge and measure the slope angle at the left corner of each triangle. Write the number on the triangle.
3. Roll each one into a cone and tape it closed.
4. Decide which is the steep volcano and which is the gentle one. Which angle number is bigger?
5. Write one sentence explaining how you knew.

Why: Thick, sticky lava piles up close to the vent and makes steep slopes, while runny lava spreads far and makes gentle ones.

🌱 **Make it Easier:** Skip the protractor. Just compare the two triangles by laying one on top of the other and saying which corner is sharper.

🚀 **Level it Up:** Cut a third triangle with exactly 45° and then one with 20°. Predict which volcano type each would be before you measure.

📚 LANGUAGE ARTS

Words, stories and headlines are the tools we use to capture a mountain that breathes fire.

📖 Challenge 1: The Volcano Dictionary

🎯 Word Origins

Where did the word "volcano" come from, and who was it named after? ❓

You're the dictionary maker, and your first entry is the word "volcano" itself. Roman myth told of Vulcan, god of fire and the forge, whose workshop was said to lie under a mountain. The word "volcano" comes from his name, from the island Vulcano near Sicily.

1. Fold 3 sheets of paper in half and staple the fold to make a little book.
2. On page one, write "volcano" and "Vulcan" and draw the god at his forge.
3. Give each new word its own page: magma, crater, caldera and ash.
4. Write each definition in your own words and draw a picture that explains it.
5. Add a cover and a title.

🌱 **Make it Easier:** Do only 3 words, and draw the picture first, then write one short sentence.

🚀 **Level it Up:** Add a "word detective" line to each page: where do you think the word comes from, and what other words does it remind you of?

📰 Challenge 2: Extra! Extra! The Mountain Blew Its Top

🎯 News Writing

What would a newspaper say on the very day a mountain lost its top? ❓

You're the newspaper editor, and it's the morning of May 18, 1980. Mount St. Helens in Washington State, USA, has erupted, and its top collapsed in a huge landslide, lowering the summit by about 400 m. Your readers need the news now!

1. Write a big headline in capital letters, short enough to shout.
2. Under it, write 3 or 4 sentences answering who, what, where, when and why.
3. Invent a quote from a witness who saw it happen, and put it in quotation marks with their name.
4. Draw a map sketch of the blast zone with the mountain, the landslide path and a label for where the ash fell.
5. Give your newspaper a name and a date at the top.

🌱 **Make it Easier:** Write only the headline and one sentence, then add the map sketch.

🚀 **Level it Up:** Add a second article on the same page, such as a report from a rescue pilot or a letter from a worried neighbor, written in a different voice.

🔦 Challenge 3: The Legend of the Smoking Mountain

🎯 Storytelling

Why do you think people long ago believed a mountain could smoke? ❓

Long before science, people told stories to explain the smoking mountains around them. Now it's your turn to invent one.

1. Decide who or what lives inside your mountain: a giant, a sleeping dragon, a grumpy chef, a lonely spirit?
2. Plan your legend on a sheet of paper with a beginning, a middle (the problem) and an ending that explains why smoke rises.
3. Practice reading it aloud 2 or 3 times. Try a slow, mysterious voice, a loud voice for the eruption and a whisper for the ending.
4. Turn off the lights, sit with your family in a circle and place a flashlight in the middle as your "campfire."
5. Tell your legend like a real storyteller, using your hands and face.

🌱 **Make it Easier:** Tell your legend in just 3 sentences, one for the beginning, one for the middle and one for the ending.

🚀 **Level it Up:** Add a repeating line, like a chant, that your listeners say with you each time the mountain rumbles.

🎨 ART & CREATIVITY

Volcanoes are some of the most dramatic things on Earth, and artists have been capturing their fire, color and smoke for thousands of years.

🌋 Challenge 1: Night Eruption in Oil Pastel

🎯 Color and Contrast

Why does a glowing color look so much brighter when it sits next to darkness? ❓

In 1991, Mount Pinatubo in the Philippines blasted an ash cloud about 35 km into the sky, and the sulfur gas it released cooled the whole planet by around 0.5°C for a year or two. Now draw a night eruption of your own.

1. Tape a sheet of black paper to a table.
2. With oil pastels, press hard to draw a mountain in dark purple or dark blue, then a tall, wide ash plume rising above it.
3. Fill the crater and the lava streams with bright red, orange and yellow. Layer the colors on top of each other.
4. Dip an old toothbrush in white or yellow paint thinned with a little water. Hold it over the paper and flick the bristles with your thumb to spray glowing sparks into the sky.

Why: dark paper makes bright colors look like they are giving off light, because your eyes compare them to the black around them.

🌱 **Make it Easier:** Skip the toothbrush and make sparks by dotting yellow pastel or a dab of paint with a fingertip.

🚀 **Level it Up:** Make a second picture of the same volcano in daylight, then compare how the colors and mood change.

🌀 Challenge 2: Marbled Magma

🎯 Printmaking

Can swirling colors on a tray of foam show how molten rock folds and flows? ❓

Molten rock swirls and folds as it flows. Capture that on paper with marbling.

1. Squirt a layer of shaving cream into a baking tray and smooth it flat.
2. Drip red, orange and yellow food coloring on top, a few drops of each.
3. Drag a toothpick or fork through the colors to make swirls.
4. Press a sheet of paper onto the foam, lift it, then scrape the cream off with a ruler to reveal the pattern.
5. Repeat with different color mixes and swirl patterns. No two prints will ever match.
6. When the prints are dry, cut your best ones into flame and lava shapes. Layer them into a glowing eruption picture and add a title card.

🌱 **Make it Easier:** Make just two prints using only two colors, and tear the shapes with your hands instead of cutting them.

🚀 **Level it Up:** Add a third layer by gluing the marbled shapes at different heights with folded paper tabs, so your eruption pops out in 3D.

🏺 Challenge 3: Curator of the Clay Blade

🎯 Sculpting and Display

How do you make something you craft yourself look like it belongs in a museum? ❓

You're the museum curator, and a new exhibit needs a star piece: a sharp, shiny, black volcanic-glass blade.

1. Mix salt dough (1 cup flour, 1/2 cup salt, 1/2 cup water) or grab air-dry clay.
2. Shape a flat, leaf-shaped blade with a pointed tip. Press a pencil along the edges to make small chips, like a real flaked tool.
3. Carve a few lines or patterns into the surface with a toothpick.
4. Let it dry fully (salt dough can take a day or two), then paint it glossy black or dark green. Add a tiny bit of white paint dry-brushed on the edges for a worn look.
5. Write a museum label with a name for the piece, where it was "found," and a one-line description. Stand the label up next to the blade.

🌱 **Make it Easier:** Skip the chipped edges and carve just one simple pattern.

🚀 **Level it Up:** Make a second blade in a different style, then write a label comparing how the two tools could have been used.

🧰 LIFE SKILLS & SEL

When a volcano wakes up, the people who stay calm and prepared are the ones who stay safe, and you can practice that now.

🎒 Challenge 1: Pack the Ash-Ready Go-Bag

🎯 Emergency Planning

What would you need if you had to leave home in ten minutes and could only carry one bag? ❓

Volcanic ash isn't soft like fireplace ash. It is made of tiny, sharp bits of rock and glass that scratch eyes and make breathing hard.

1. On paper, write a go-bag checklist for your family: water, snacks, flashlight, a first-aid kit, a family phone list, copies of important papers, and one comfort item.
2. Add the things only your family needs, like medicine, glasses, or pet food.
3. Hold a dust mask, or a clean cloth dampened with water, over your nose and mouth. Walk across the room and climb a few stairs. How does breathing feel?
4. Gather what you can from around the house and pack your go-bag, checking each item off your list.

Why: a damp cloth traps tiny particles better than dry fabric, but it also makes breathing a bit harder, which is why you test it before you need it.

🌱 **Make it Easier:** Pick only 5 items for your checklist and draw a picture of each one instead of writing.

🚀 **Level it Up:** Write what each item is for in one sentence, then ask a family member to find one thing you forgot.

🚨 Challenge 2: Evacuation Drill at Taal

🎯 Planning Under Pressure

How fast could your family get from "warning!" to "everyone is safe"? ❓

You are the emergency coordinator. Imagine officials near Taal Volcano, a volcano sitting inside a lake in the Philippines, have just announced a warning, and your family must leave.

1. Choose two meeting points: one close to home and one farther away. Draw a simple map showing the routes to both.
2. Decide who calls whom, and write down 3 phone numbers everyone should know.
3. Set a timer and shout "Warning!" Everyone grabs their go-bag (or pretend bag), puts on shoes, and walks to the meeting point.
4. Stop the timer and check that everyone made it. What slowed you down?
5. Fix one problem and run the drill again. Can you beat your time?

Why: practicing a plan in advance helps people stay calm, because their bodies already know what to do when stress makes thinking harder.

🌱 **Make it Easier:** Practice with only one meeting point and skip the phone calls.

🚀 **Level it Up:** Run the drill again in the dark with only a flashlight, or pretend one exit is blocked and find a new route.

🌡️ Challenge 3: Feelings Thermometer

🎯 Managing Big Feelings

What do you do with your body and mind when something in nature feels scary? ❓

People facing an emergency often feel worried, shaky, angry, or sad, and all of those feelings are normal.

1. Draw a tall thermometer on paper and divide it into 5 levels, from "Calm" at the bottom to "Very Scared" at the top.
2. Color each level and give it a feeling word and a face.
3. Beside each level, list what your body feels like there, such as a fast heartbeat, tight shoulders, or a loud voice.
4. Try these calming tools and rate each one: belly breathing (inhale for 4 counts, exhale for 6), a big hug, squeezing a pillow, or thinking of how you could help someone else.
5. Next to the top levels, write the tool that works best for you.

🌱 **Make it Easier:** Use only 3 levels (calm, worried, scared) and pick just one calming tool.

🚀 **Level it Up:** Interview a family member about a time they felt scared, then add their calming tool to your list and try it together.

🏃 MOVEMENT & BODY

Volcanoes warn us with shaking ground, so let's train your body and your brain to feel, read and escape the rumble.

🌋 Challenge 1: Be the Seismometer

🎯 Balance and Recording

How can a shaking line on paper tell you whether someone is tiptoeing or stomping?❓

Volcanologists use seismometers, which record ground shaking, to watch for signs of an eruption. Now it's your turn.

1. Tape a cardboard box to the floor, open side down. Poke a hole in the top, hang a pen from a string through the hole so it dangles over the edge, and let its tip just touch a long paper strip underneath.
2. Ask a friend to slide the strip slowly under the pen while you stand nearby.
3. Have them tiptoe, walk, jump and stomp, close to the box and then far from it. Label each wiggly line on the strip.
4. Now become your own seismometer. Stand on one foot on a cushion while someone stomps next to you. Count how many seconds you stay balanced, and record it for each round.
5. Practice "drop, cover and hold on": get low, cover your head and neck, and grab something sturdy.

Why: the box shakes but the hanging pen wants to stay still, so the paper moves under it and draws the shaking.

🌱 **Make it Easier:** Skip the balance test and just compare the lines for tiptoe and stomp.

🚀 **Level it Up:** Repeat the stomps from 1, 2 and 3 steps away. Does the line get smaller with distance? Do the same for your balance times.

🏃 Challenge 2: Lahar Escape!

🎯 Running and Agility

Why do people escape a river of mud by climbing to high ground instead of running down the valley? ❓

Build an escape course in a living room or yard. Roll a blanket into a long log: that's your "mudflow." Set up cushions to climb over and a "safe ground" spot at the top, such as a stack of pillows or the top step.

1. Players line up at the bottom, with the mudflow rolled out behind them.
2. A caller shouts "Lahar!" and everyone runs uphill (or toward the safe spot), climbing over the cushions.
3. The caller pushes the blanket roll slowly after the runners, who must reach safe ground before it touches them.
4. Players who make it earn a point. Keep a scoreboard on paper and play five rounds, making the course harder each time.

🌱 **Make it Easier:** Play in a flat, open space and let the mudflow start far behind the runners.

🚀 **Level it Up:** Add a rule that you must carry a "go bag" (a stuffed animal or small backpack) the whole way, and time each round with a stopwatch.

🚀 ASTRONOMY

Volcanoes aren't only on Earth: the biggest and the busiest ones in the solar system are on other worlds.

🏔️ Challenge 1: Stretch Out Olympus Mons

🎯 Measuring and Comparing

If a volcano on Mars poked up through your living room, how far above Earth's tallest mountain would it reach? ❓

You're the space surveyor, and your job is to measure a mountain that no human has ever touched.

1. Look up Mount Everest's height above sea level: about 8.8 km. Olympus Mons on Mars is about 22 km high.
2. Pick a scale: 1 cm of tape for every 1 km. Everest becomes about 9 cm of tape.
3. Stretch a tape measure or long string along the floor and mark 9 cm. Label it "Everest" with a sticky note.
4. Keep going to 22 cm and label it "Olympus Mons."
5. Measure the gap between the two labels. How many centimeters is it, and how many kilometers does that stand for?
6. Add a third mountain you know, like a hill near your home, to your line.

Why: Mars has a thin crust that doesn't move in big plates like Earth's, so a volcano can keep building in one spot for a very long time and grow far taller than any volcano on Earth.

🌱 **Make it Easier:** Use a scale of 1 cm for every 2 km, so your numbers are smaller and round: Everest is about 4 cm and Olympus Mons is 11 cm.

🚀 **Level it Up:** Rebuild your line at a scale of 1 meter for every 1 km. Stretch it across a hallway or yard, and walk along the full 22 meters.

🌋 Challenge 2: Postcard from Io

🎯 Imaginative Design

What would it feel like to visit a place where volcanoes erupt almost all the time? ❓

You're the alien tour guide for Io, a moon of Jupiter with hundreds of volcanoes, the most volcanically active place in our solar system.

1. Fold a sheet of paper in half to make a postcard.
2. On the front, draw what a visitor would see from a spaceship window: erupting volcanoes, a giant Jupiter filling the sky, and a yellow-orange surface.
3. On the back, write a short message to a friend back home. Describe what you hear, what you feel under your feet, and what makes you nervous.
4. Add a travel tip for future visitors, such as "Never camp near a fresh eruption!"
5. Choose a tourist slogan, like "Come see a volcano that never takes a day off!"
6. Copy your drawing and slogan onto a big sheet of paper, then add a title and eruption plumes to make it a poster.

🌱 **Make it Easier:** Skip the poster. Draw only the postcard front, and write two short sentences on the back.

🚀 **Level it Up:** Add a price list for your tour with at least three activities, like "Eruption Watching," and make up a price for each one. Work out the total cost for a family of four.